Algorithms’ Project Description

Overview:

This project is designed to immerse you in the practical aspects of computer science research, focusing on the implementation and evaluation of a specific algorithm presented in an assigned research paper (Each team is assigned to a Research paper randomly). You will work in teams to dive deep into the paper, understand the presented algorithm, and implement it.

The ultimate goal is to replicate the results presented in the paper, especially in terms of performance metrics against the baselines presented in the paper.

<u>What do we mean by baseline:</u> For example, if the paper is proposing algorithm called Super-Merge-Sort, in the paper you will see it is compared against a few other basic algorithms, in this example it will be the classical Merge Sort.

Objectives:
<u>Comprehension:</u> Develop a thorough understanding of the assigned research paper, with an emphasis on the proposed algorithm, and the evaluation methodology.

<u>Implementation:</u> Translate the pseudocode or algorithmic descriptions from the paper into a working program. Your code should be clean, well-documented, and structured in a way that reflects the logic and flow of the original algorithm in Java Language.

<u>Evaluation:</u> Replicate the evaluation process as described in the paper. This involves generating or using the specified datasets, running your implementation under the defined conditions (as close as possible), and comparing your results with the baseline results presented in the research.

(<u>**Note:**</u> for some of the assigned papers evaluation under the exact conditions set in the paper is not possible, in these case we expect you to evaluate it in conditions as close as possible to the original paper, but explain it to us why it was not possible to follow same process in the final report!)

<u>Analysis:</u> Critically analyze the performance of your implementation. Discuss any deviations from the expected results, potential reasons for these discrepancies, and any insights gained during the implementation process.

Deliverables:
Below you will see the list of deliverables for this project. All these must be submitted on BrightSpace, at the defined deadlines.

Role Assignment (Submission Deadline: **Due Week 9):** We need you to submit an exact definition of the responsibilities of each team member. The potential roles can include: Developer, Analyser, Communicator, Documenter. This does not mean that each person should only take 1 role. We would ideally want each of you to cooperate in all tasks and only have one person as the lead for each of the roles mentioned above.

For example: Lucy and Sam can both work on the development, but Lucy could take the lead.

Or when the implementation is heavy, you could split between two teams of developers, and one lead for each team! This is something we need you to discuss amongst your team and make a decision!

A google Form will be distributed for this deliverable!

Project Report (Submission Deadline: End of Week 12): A comprehensive document (3-page max) detailing your understanding of the paper, the implementation process, challenges faced, and the results of your evaluation. Include a section comparing your results with those in the paper and discuss any variances. (**Note:** A template for the report is available on Bright Space):

Code (Submission Deadline: End of Week 12): A complete set of source code files for the implemented algorithm, accompanied by a README file that includes instructions for running the code and reproducing your results.
Your code should be well-commented to explain critical sections and the logic behind your
implementation choices.

Presentation (Submission Deadline: Week 11): Prepare a poster summarizing your project, highlighting key aspects of the paper, your implementation journey, results, and learnings. This presentation will be delivered in a poster presentation event, followed by a Q&A segment. (Note: A template for the poster is also available on BrighSpace).

Evaluation Criteria and Rubric

Accuracy of Implementation: Does the implemented algorithm accurately reflect the pseudocode or descriptions provided in the paper? (30%)

Results Replication: How closely do your results match those presented in the paper in terms of graphical outputs and performance metrics? (20%)

Understanding and Analysis: Demonstration of a deep understanding of the algorithm and the ability to critically analyze and discuss the implementation and its outcomes.(20%)

Documentation and Presentation: Quality of the project report, clarity of the code documentation, and effectiveness of the presentation in conveying your project's scope, challenges, and achievements. (30%)

Teamwork and Collaboration:

Teamwork is an important aspect of this project. You are expected to communicate with your team members, have a team lead and communication plans in place, and a clear task breakdown for the implementation and presentation of your algorithm. Pair programming can be a very useful practice.

Regular meetings should be scheduled to discuss the paper, divide tasks, and monitor progress. Collaboration tools (e.g., Git) are recommended for coordinating work and tracking
changes.

Assignment of Papers to Groups:

Group 1 – 9 Paper 1
Group 10 – 18 Paper 2
Group 19 – 27 Paper 3
Groups 28 – 36 Paper 4